

Mechanics Tutorial 01- Answers.

①

$$\underline{v}_1 = 8\underline{i} + 12\underline{j} \text{ m s}^{-1} \text{ at } t = 0.5$$

$$\underline{v}_2 = 6\underline{i} + (-5)\underline{j} \text{ m s}^{-1} \text{ at } t = 2.0$$

assumption of constant acceleration.

$$\underline{a} = a_x \underline{i} + a_y \underline{j} \text{ is taken as acceleration vector.}$$

apply $\underline{v} = \underline{u} + \underline{a}t$ $\underline{i}, \underline{j}$ directions separately $\underline{i} \rightarrow$

$$\underline{v} = \underline{u} + \underline{a}t$$

$$6\underline{i} = 8\underline{i} + a_x \times 2.0$$

$$a_x = -0.1 \underline{i} \text{ m s}^{-2}$$

 $\underline{j} \rightarrow$

$$-5\underline{j} = 12\underline{j} + a_y 2.0$$

$$a_y = -0.85 \underline{j}$$

$$\therefore \underline{a} = -0.1\underline{i} - 0.85\underline{j} \text{ m s}^{-2}$$

Note

if equations are applied in $\underline{i}, \underline{j}$ directions separately no need to write $\underline{i}$ and $\underline{j}$ unit vector during calculation. in final answer it is must to indicate $\underline{i}$ and $\underline{j}$

②

$$\underline{x}_0 = 0\underline{i} + 0\underline{j} \text{ at } t = 0.5$$

(0,0) is the origin

$$\underline{v}_0 = 8\underline{i} + 5\underline{j} \text{ m s}^{-1}$$

$$\underline{a} = 0.2\underline{i} + 0.8\underline{j} \text{ m s}^{-2}$$

at t

$$\underline{x}_t = p_x \underline{i} + p_y \underline{j} \text{ ?}$$

$$\underline{v}_t = v_x \underline{i} + v_y \underline{j} \text{ ?}$$

at $t = t$ we take the components as follows;

$$\underline{x}_t = p_x \underline{i} + p_y \underline{j}$$

$$\underline{v}_t = v_x \underline{i} + v_y \underline{j}$$

apply $\underline{v} = \underline{u} + \underline{a}t$ in $\underline{i}$ dirⁿ

$$\underline{v} = \underline{u} + \underline{a}t$$

$$v_x = 8 + 0.2 \times t$$

$$v_x = (8 + 0.2t) \underline{i}$$

apply $\underline{v} = \underline{u} + \underline{a}t$ in $\underline{j}$ dirⁿ

$$v_y = 5 + 0.8t \underline{j} \Rightarrow v_y = (5 + 0.8t) \underline{j}$$

$$\therefore \underline{v}_t = (8 + 0.2t) \underline{i} + (5 + 0.8t) \underline{j}$$

this is the velocity vector at $t = t$

$t = t$ in $\underline{i}$ dirⁿ ~~position vector~~ displacement $\underline{s}_x$

$$\underline{s}_x = \underline{p}_x \underline{i} - 0 \underline{i}$$

$\underline{R}$ this is from position vector at $t = 0$ $\underline{p}$

$$\therefore \underline{s}_x = \underline{p}_x \underline{i}$$

$t = t$ in $\underline{j}$ dirⁿ displacement $\underline{s}_y$

$$\underline{s}_y = \underline{p}_y \underline{j} - 0 \underline{j}$$

$$\therefore \underline{s}_y = \underline{p}_y \underline{j}$$

apply $s = ut + \frac{1}{2}at^2$ $\underline{i}$ dirⁿ

$$\underline{p}_x = 8 \times t + \frac{1}{2} \times 0.2 t^2$$

$$\underline{p}_x = (8t + 0.1t^2) \underline{i}$$

apply $s = ut + \frac{1}{2}at^2$ $\underline{j}$ dirⁿ

$$\underline{s}_y = \underline{p}_y = 5 \times t + \frac{1}{2} \times 0.8 t^2$$

$$\underline{p}_y = (5t + 0.4t^2) \underline{j}$$

$$\therefore \underline{s}_t = \underline{p}_x \underline{i} + \underline{p}_y \underline{j}$$

$$= (8t + 0.1t^2) \underline{i} + (5t + 0.4t^2) \underline{j}$$

③.

$$\underline{v}_0 = (8\underline{i} + \underline{j}) \text{ m s}^{-1}$$

$$\underline{x}_0 = (0\underline{i} + 1.5\underline{j}) \text{ m}$$

$$\underline{a} = 0\underline{i} - 10\underline{j} \text{ m s}^{-2}$$

(a). $\underline{x}_t$? at $t = t_f$ $\underline{x}_t = P_x \underline{i} + P_y \underline{j}$

apply $s = ut + \frac{1}{2}at^2$ in $\underline{i}$ dirⁿ

$\underline{s}$ = displacement vector at time $t = t$ & $t = 0$

$$\underline{s} = \underline{x}_t - \underline{x}_0$$

$$= (P_x \underline{i} + P_y \underline{j}) - (0\underline{i} + 1.5\underline{j})$$

$$\underline{s} = P_x \underline{i} + (P_y - 1.5) \underline{j}$$

& this a vector which gives displacement during the time period

apply $s = ut + \frac{1}{2}at^2$ in $\underline{i}$ dirⁿ

$$P_x = 8 \times t + \frac{1}{2} \times 0 \times t^2 = 8t$$

apply $s = ut + \frac{1}{2}at^2$ in $\underline{j}$ dirⁿ

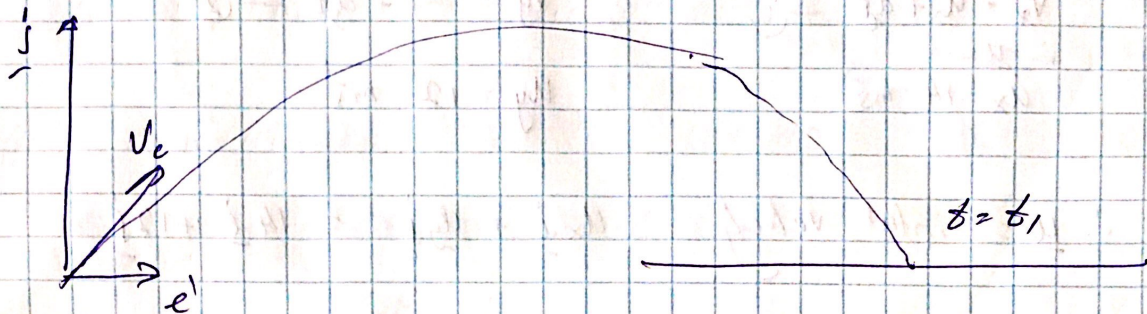
$$P_y - 1.5 = 1 \times t + \frac{1}{2} \times (-10) t^2$$

$$P_y = 1.5 + t - 5t^2$$

$$\therefore \underline{x}_t = P_x \underline{i} + P_y \underline{j} = 8t \underline{i} + (1.5 + t - 5t^2) \underline{j}$$

(b). the position vector of the ball at any time, $t \geq 0$ is given as,

$$\underline{r}_t = 8t\underline{i} + (1.5 + t - 5t^2)\underline{j}$$



when $t = t_1$ the ball hits on the ground. So the vertical component ($\underline{j}$ dirⁿ) should be zero i.e.

$\underline{r}_t$ in $\underline{j}$ dirⁿ make zero.

$$\therefore 1.5 + t_1 - 5t_1^2 = 0$$

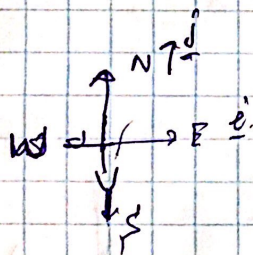
$$5t_1^2 - t_1 - 1.5 = 0$$

$$\therefore t_1 = 0.656 \text{ s} \text{ or } t_1 = -0.456 \text{ s}$$

$$\therefore t_1 = 0.656 \text{ s}$$

t is not negative

(4).



directors

$$t = 0 \text{ s} \quad \underline{r}_0 = 0\underline{i} + 0\underline{j} \text{ origin.}$$

$$t = t_1 \text{ s} \quad \underline{r}_{t_1} = 2(7-t_1)\underline{i} + 4(3-t_1)\underline{j}$$

(a). if ball is heading south $\underline{v}_{t_1} = v_x\underline{i} + v_y\underline{j}$

i.e. horizontal component $v_x = 0$ and $v_y < 0$

$$\therefore \underline{v}_{t_1} = \underbrace{2(7-t_1)}_{v_x}\underline{i} + \underbrace{4(3-t_1)}_{v_y}\underline{j}$$

$$v_x = 0 = 2(7-t_1)$$

$$2 \times 7 = t_1 = 7 \text{ s}$$

$$v_y < 0$$

$$3 - t_1 < 0$$

$$\therefore t_1 > 3 \text{ s}$$

$\therefore t_1 = 7 \text{ s}$ satisfies both conditions.

(b).

$$v_x = 2(7-t)$$

$$v_y = 4(3-t)$$

$$v_x = 14 + (-2)t$$

$$v_y = 12 + (-4)t$$

$$v_x = u_x + at \quad \text{--- (1)}$$

$$v_y = u_y + at \quad \text{--- (2)}$$

$$u_x = 14 \text{ m/s}$$

$$u_y = 12 \text{ m/s}$$

(i) $\therefore \underline{u} = \text{initial velocity} = u_x \underline{i} + u_y \underline{j} = 14 \underline{i} + 12 \underline{j}$

(ii) from (1) and (2)

$$a_x = (-2) \text{ m/s}^2$$

$$a_y = -4 \text{ m/s}^2$$

$$\underline{a} = a_x \underline{i} + a_y \underline{j} = -2 \underline{i} - 4 \underline{j} \text{ m/s}^2$$

(c). distance between origin = $\underline{s} = s_x \underline{i} + s_y \underline{j}$ at $t = 10 \text{ s}$

apply $s = ut + \frac{1}{2}at^2 \rightarrow \underline{s} \text{ dir}^n$

$$s_x = 14 \times 10 + \frac{1}{2} \times (-2) \times 10^2$$

$$= 40 \text{ m}$$

apply $s = ut + \frac{1}{2}at^2 \quad \underline{j} \text{ dir}^n$

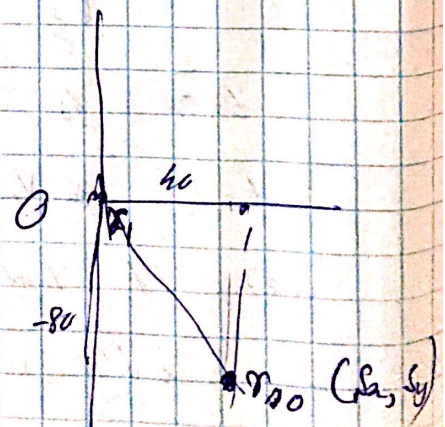
$$s_y = 12 \times 10 + \frac{1}{2} \times (-4) \times 10^2$$

$$= -80 \text{ m}$$

$$\therefore \text{distance vector} = \underline{s} = 40 \underline{i} - 80 \underline{j}$$

$$\therefore \text{Distance between two points} = |\underline{s}| = \sqrt{40^2 + 80^2}$$

$$\text{Distn origin and point at } t=10 \text{ s} = 89.44 \text{ m}$$



(5)

$$\underline{a} = 0.9\underline{i} + 0.7\underline{j}$$

$$\underline{v}_0 = 4\underline{i} + 0\underline{j}$$

$$\underline{r}_0 = 400\underline{i} + 350\underline{j}$$

(a). apply $\underline{v} = \underline{u} + \underline{a}t$ $\underline{i}$ dir $t = 10s$

$$v_x = 4 + \frac{1}{2} \times 0.9 \times 10 = 13 \text{ ms}^{-1}$$

apply $\underline{v} = \underline{u} + \underline{a}t$ $\underline{j}$ dir

$$v_y = 0 + 0.7 \times 10 = 7 \text{ ms}^{-1}$$

$$\underline{v}_{10} = 13\underline{i} + 7\underline{j} \text{ ms}^{-1}$$

(b).

$$\underline{s} = \text{displacement vector} = \underline{r}_{10} - \underline{r}_0 \Rightarrow \underline{s} = s_x \underline{i} + s_y \underline{j}$$

$$\underline{s} = s_x \underline{i} + s_y \underline{j}$$

$$\underline{r}_{10} = r_x \underline{i} + r_y \underline{j}$$

$$\underline{s} = \underbrace{(r_x - 400)}_{s_x} \underline{i} + \underbrace{(r_y - 350)}_{s_y} \underline{j}$$

$$s = \underline{u} + \frac{1}{2} \underline{a} t^2 \Rightarrow \underline{i} \text{ dir } t = 10s$$

$$r_x - 400 = 4 \times 10 + \frac{1}{2} \times 0.9 \times 10^2$$

$$\therefore r_x = 485 \text{ m}$$

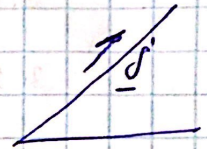
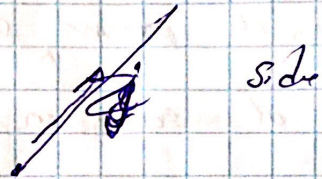
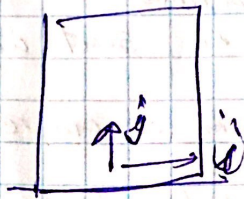
$$s = \underline{u} + \frac{1}{2} \underline{a} t^2 \quad \underline{j} \text{ dir}$$

$$r_y - 350 = 0 \times 10 + \frac{1}{2} \times 0.7 \times 10^2$$

$$r_y = 385 \text{ m}$$

$$\therefore \underline{r}_{10} = 485\underline{i} + 385\underline{j}$$

$\hat{i}$ dirⁿ & $\hat{j}$ dirⁿ along the plane



or diff

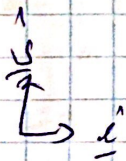
$$\underline{v}_0 = 5\hat{i}, \quad \underline{a} = -4\hat{j}, \quad \underline{s}_0 =$$

(a) when $t = 4s$

* apply $v = u + at$ in $\hat{i}, \hat{j}$ dirⁿ separately
get the velocity vector from

(b) apply $s = ut + \frac{1}{2}at^2$ at $t = 4s$ or $\hat{i}, \hat{j}$ dirⁿ separately. $\underline{s}$ displacement should be defined.

⑦.



$$\underline{a} = \text{acceleration vector} = 2\hat{i} + 3\hat{j}$$

$$v_0 = 0$$

$$t = 4s$$

$$\underline{v}_4 = ?$$

$$\underline{s}_0 = 0\hat{i} + 0\hat{j}$$

$$\underline{s}_4 = ?$$

apply $v = u + at$ on $\hat{i}$ and $\hat{j}$ separately and get $\underline{v}_4$
the velocity vector at $t = 4s$

$$\underline{s} = \text{displacement vector from } t=0 \text{ to } t=4s \quad \underline{s} = \underline{s}_4 - \underline{s}_0 \quad \text{--- (1)}$$

then apply $s = ut + \frac{1}{2}at^2$ on $\hat{i}, \hat{j}$ dirⁿ separately and
substitute on (1) equation get $\underline{s}_4$ (the position vector)

(8).

$$\underline{a} = 0.2\underline{i} + 0.3\underline{j} \text{ m/s}^2$$

$$\underline{V}_0 = 4\underline{i} - 8\underline{j} \text{ m/s}$$

$$\underline{r}_0 = 6\underline{i} + 2\underline{j} \text{ m}$$

(a) at $t = 10 \text{ s}$ $\underline{V}_{10} = ?$
 apply $\underline{v} = \underline{u} + \underline{a}t$ both $\underline{i}$ and $\underline{j}$ dirⁿ get $\underline{V}_{10}$ vector

(b) at $t = 10 \text{ s}$ position vector $\underline{r}_{10} = r_{10x}\underline{i} + r_{10y}\underline{j}$
 apply $\underline{r} = \underline{u}t + \frac{1}{2}\underline{a}t^2$ in both $\underline{i}$ and $\underline{j}$ dirⁿ for
 displacement vector $\underline{r} = \underline{r}_{10} - \underline{r}_0$

(c) $\underline{V}_t = 4.8\underline{i} - 6.8\underline{j}$ $t = t \text{ s}$

$\underline{v} = \underline{u} + \underline{a}t$ $\underline{i}$ dirⁿ $\underline{j}$ dirⁿ

$$4.8 = 4 + 0.2 \times t$$

$$-6.8 = -8 + 0.3 \times t$$

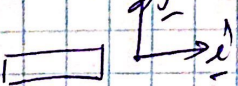
$$t = 4 \text{ s}$$

$$t = 4 \text{ s}$$

both gives $t = 4 \text{ s}$

(9).

$$t = 0 \text{ s}$$



$$\underline{V}_0 = 80\underline{i} \text{ m/s}$$

$$\underline{a}_0 = \underline{i} + 4\underline{j} \text{ m/s}^2 \text{ for } 30 \text{ s}$$

(a) at $t = t \text{ s}$

$$\underline{V}_t = V_{tx}\underline{i} + V_{ty}\underline{j}$$

Atlas

apply v initial $\underline{i}$ dir $\hat{i}$

$$v_x = 80 + \frac{1}{2} \times t$$

$$v_x = 80 + t$$

$$\therefore \underline{v}_t = (80 + t)\underline{i} + 4t\underline{j} \quad \text{velocity vector at any } t = t'$$

$$\underline{r}_0 = 0\underline{i} + 0\underline{j}$$

$\underline{r}$ displac vect. for $t=0$ to $t=t'$

$$\underline{r} = \underline{r}_t - \underline{r}_0$$

$$\underline{r} = \underline{r}_t - \underline{r}_0$$

$\underline{r}_t =$ posn vect. at $t=t'$

$$\underline{r} = p_x \underline{i} + p_y \underline{j}$$

$$\underline{r}_t = p_x \underline{i} + p_y \underline{j}$$

$$\underline{r}_t = p_x \underline{i} + p_y \underline{j}$$

apply $\underline{r}$ at $\Rightarrow \underline{i}$, $\underline{j}$ dir

$$p_x = 80 \times t + \frac{1}{2} \times 1 \times t^2$$

$$p_x = 80t + \frac{1}{2}t^2$$

or $\underline{j}$ dir $s = ut + \frac{1}{2}at^2$

$$p_y = 0 + \frac{1}{2} \times 4t^2$$

$$p_y = 2t^2$$

$$\therefore \underline{r}_t = (80t + \frac{1}{2}t^2)\underline{i} + 2t^2\underline{j} \quad \text{--- (1)}$$

(b), at height is 50m the $\underline{j}$ dir comp of above

$\underline{r}_t$ should be 50m

$$\therefore 2t^2 = 50$$

from (1) substitute eqn $\underline{j}$ comp to eqn

$$t = \sqrt{25} \quad t = 5s$$

(10)

$$\underline{v}_0 = \underline{i} + 2\underline{j} \text{ ms}^{-1} \quad \underline{a} = \underline{f} = a_x \underline{i} + a_y \underline{j} \text{ can be taken}$$

(a). apply $v = u + at$ on $\underline{i}$ and $\underline{j}$ dirⁿ separately.
for $t = 10\text{s}$
get a_x and a_y and substitute to get $\underline{f}$
the acceleration vector.

(b). simply find position vector at $t = 50\text{s}$. consider
start at $t = 0$ separately from $t = 0 \rightarrow t = 10\text{s}$ and
then from $t = 10\text{s}$ to $t = 50\text{s}$

(11)

$$\underline{v}_0 = 8\underline{i} + 10\underline{j} \text{ ms}^{-1}$$

$$\underline{a} = -10\underline{j} \text{ ms}^{-2}$$

$$\underline{r}_0 = 0\underline{i} + 10\underline{j}$$

(a). at $t = T\text{s}$ ball hits the ground so, its position
vector at $t = T\text{s}$ is $\underline{r}_T = R_T \underline{i} + R_T \underline{j}$ the $\underline{j}$ dirⁿ
component (R_T) should be zero
 $R_T = 0$ ①

apply $s = ut + \frac{1}{2}at^2$ in $\underline{j}$ dir $\uparrow$

$$R_T = 0 = 10 \times T + \frac{1}{2} \times (-10) T^2$$

from ① $10T + (-5)T^2 = 0$

$$\Rightarrow T = 0 \quad -5T + 10 = 0$$

$$T = 2\text{s}$$

$$T = 0 \quad T = 2\text{s}$$

initial position so 1st hit is on $T = 2\text{s}$

(b) apply $s = ut + \frac{1}{2}at^2$ and get $\underline{r}_T = 2\text{s}$ so get
the $\underline{j}$ dirⁿ displacement component

(c). at maximum height $T = t\text{s}$.

at maximum height the velocity
vector $\underline{v}_b$ should have zero velocity in $\underline{j}$ dirⁿ
 ~~$\underline{v}_b = v_{bx} \underline{i} + v_{by} \underline{j}$~~ the velocity

if $\underline{v} = v_x \underline{i} + v_y \underline{j}$

$v_y = 0$ at maximum height

apply $v = u + at$ in $\underline{j}$ dirⁿ

$$v_y = 0 = 8 + \frac{1}{2} \times (-10) t$$

$$t = 1.6 \text{ s}$$

then apply $s = ut + \frac{1}{2} at^2$ in $\underline{j}$ dirⁿ.

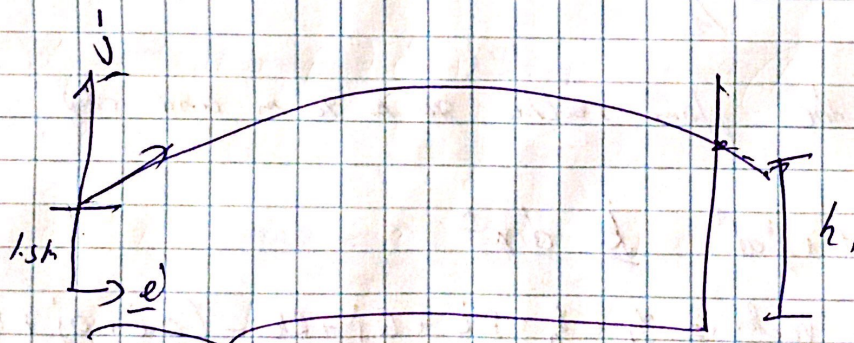
OR
apply $v^2 = u^2 + 2as$ in $\underline{j}$ dirⁿ where

$$v_y^2 = 0^2 = 10^2 + 2 \times (-10) s$$

$$s = 5 \text{ m}$$

(12). Similar questions were already discussed

(13).



for $t=0$

$$\underline{v}_0 = 15 \underline{i} + 18 \underline{j} \text{ ms}^{-1}$$

$$\underline{s}_0 = 1.5 \underline{j} \text{ m}$$

for $t=3 \text{ s}$

$$\underline{s}_3 = P_3 \underline{i} + R_3 \underline{j}$$

$$a = -10 \underline{j} \text{ ms}^{-2} \text{ is constant}$$

$$(a). \underline{s}_{0 \rightarrow 3} = \underline{s}_3 - \underline{s}_0 = (P_3 - 0) \underline{i} + (R_3 - 1.5) \underline{j}$$

$\underline{j}$ dir $\uparrow s = ut + \frac{1}{2} at^2$ $t = 3 \text{ s}$ $\rightarrow$ dir $\underline{s} = ut + \frac{1}{2} at^2$

$$R_3 - 1.5 = 18 \times 3 + \frac{1}{2} (-10) \times 3^2$$

$$R_3 = 10.5 \text{ m}$$

$$P_3 - 0 = 15 \times 3 + 0$$

$$P_3 = 45 \therefore$$

$$\underline{r}_3 = 45\underline{i} + 10.5\underline{j} \text{ m}$$

so the point the arrow hits is given by above vector

(b). apply $\underline{v} = \underline{u} + \underline{at}$ on $\underline{i}, \underline{j}$ dirⁿ separat^{ly} and
the velocity vector.

(14).

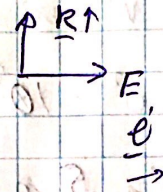
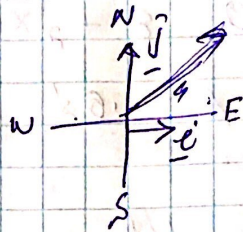
$$\underline{r}_0 = 0 + 0 + 10\underline{k} \text{ m}$$

$$\underline{v}_0 = 8\sqrt{2}\underline{i} - 8\sqrt{2}\underline{j} \text{ ms}^{-1}$$

$$\underline{r}_t = P\underline{i} + Q\underline{j} + R\underline{k}$$

So, when $R = 90 \text{ m}$ the — ①
plane stop acceleration.

$$\underline{v}_0 = 8\sqrt{2} \text{ m North East dirⁿ}$$



$$\text{So, } \underline{v}_0 = 8\sqrt{2} \times \frac{1}{\sqrt{2}} \underline{i} + 8\sqrt{2} \times \frac{1}{\sqrt{2}} \underline{j} + 0\underline{k} \quad \underline{a} = 0.01\underline{i} + 0.02\underline{j} + 0.1\underline{k}$$

$$= 8\underline{i} + 8\underline{j} \text{ ms}^{-1}$$

(a) $t = t_1$ the plane reaches ~~90 m~~ 90 m and stop
acceleration.

$$\text{apply } s = \underline{ut} + \frac{1}{2}\underline{at}^2 \text{ dirⁿ}$$

$$\underline{s} = \text{displacement vector} = \underline{r}_1 - \underline{r}_0 = P\underline{i} + Q\underline{j} + R\underline{k} - (0\underline{i} + 0\underline{j} + 10\underline{k})$$

$$= P\underline{i} + Q\underline{j} + (R - 10)\underline{k}$$

$$R - 10 = 0 + \frac{1}{2} \times 0.1 \times t^2$$

substit^{ute} from eqn^①

since it reaches 90 m
at this point,

$$90 - 10 = \frac{1}{2} \times 0.1 t^2$$

$$t^2 = 1600$$

$$t = 40 \text{ s}$$

∴ the plane accelerate for 40 s

Ans

(b) how time $t = 40s$ is known
 get apply $v = u + at$ in each $\underline{i}$, $\underline{j}$, $\underline{k}$ dirⁿ to
 get velocity component in each direction at $t = 40s$

$$\underline{V}_{40} = -\underline{i} + -\underline{j} + -\underline{k}$$

(c).

initial position of the car $\underline{r}_{C0} = 0\underline{i} + 0\underline{j}$

Since car is not moving in $\underline{k}$ dirⁿ we can
 neglect that for the car

$$\underline{r}_{C0} = 0\underline{i} + 0\underline{j} \quad \text{--- (2)}$$

when airplane stop accelⁿ. the time $t = 40s$

get the position vector of plane at $t = 40s$

$$\underline{r}_b = p\underline{i} + a\underline{j} + 90\underline{k}$$

find p and a

$\underline{i}$ dirⁿ $v = u + at$ $S = ut + \frac{1}{2}at^2$ for plane,

$$p = 8 \times 40 + \frac{1}{2} \times 0.01 \times (40)^2$$

$$= 328 \text{ m}$$

$\underline{j}$ dirⁿ $s = ut + \frac{1}{2}at^2$ for plane

$$a = 8 \times 40 + \frac{1}{2} \times 0.02 \times 40^2$$

$$= 336 \text{ m}$$

$$\therefore \underline{r}_b = 328\underline{i} + 336\underline{j} + 90\underline{k}$$

initial posⁿ $0\underline{i}, 0\underline{j}$

from (2) $\therefore \underline{r}_{C40} = p\underline{i} + a\underline{j} = 328\underline{i} + 336\underline{j}$
 $= 328\underline{i} + 336\underline{j}$

$$\text{Distance travelled} = D = \sqrt{328^2 + 336^2} \text{ m}$$